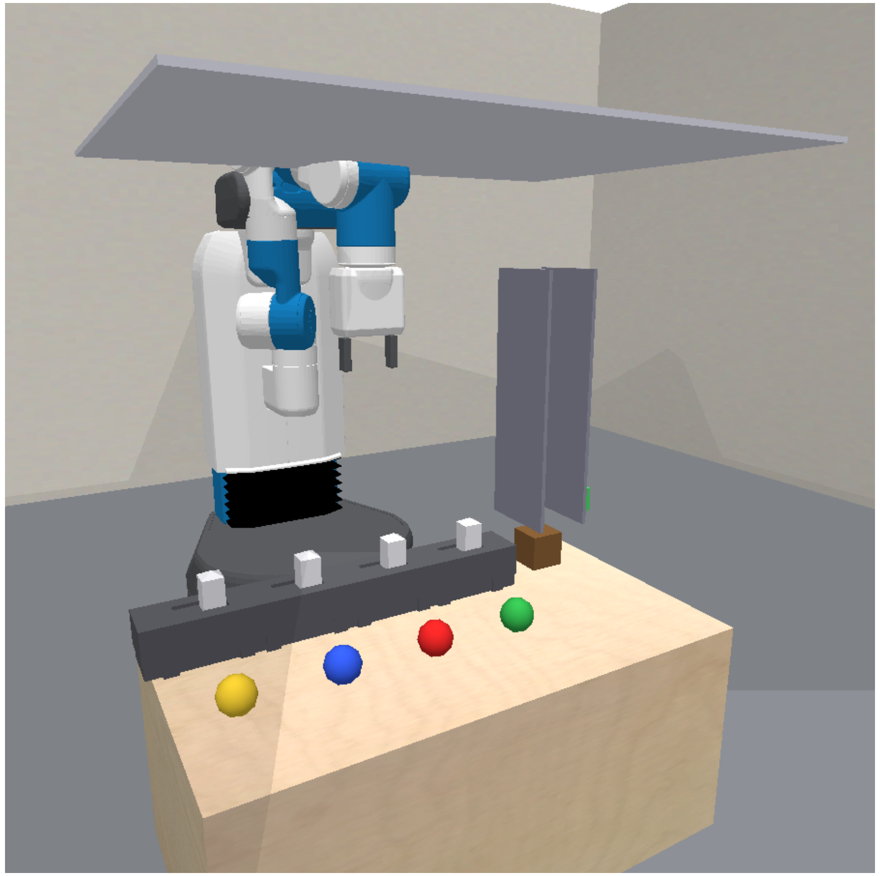
The Balloons Domain

Float a box through a chute

Choose which balloons to release so the box reaches the green height band at low speed, without bursting a balloon.

The sweep version combines hidden lift dynamics with tilt and wall contact.

September 7 uniform sweep
Seeds 0, 1, 2 · two train levels + one test level



Actual sweep frame: model-based seed 0, level 3.

VERSION USED IN THE SWEEP

What changed from the original puzzle

Transient motion

Drag is **2.2**, down from 12. The rising box can overshoot its predicted rest height.

A balloon can burst at the ceiling even when the predicted hover height is inside the band.

Tilt and contact

Balloon attachments are spread across the box top by clip index.

Off-centre pulls can tilt or sway the box into the **two chute walls**.

A jamming alternative

The test generator requires a subset that looks good by hover height but fails to settle in the band without bursting.

The intended discriminator is **contact during ascent**.

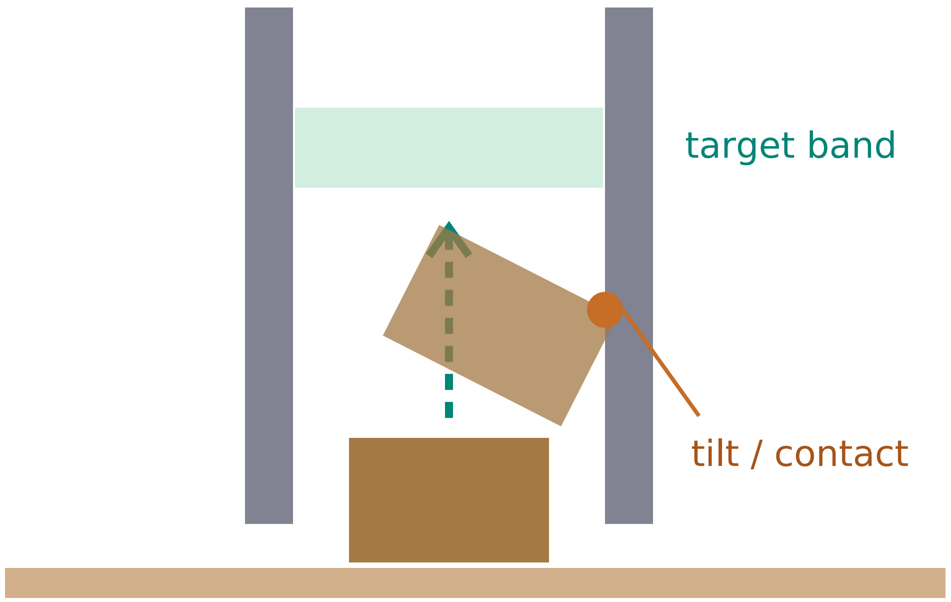
> Predicted hover height is a useful filter. It does not certify that the ascent will succeed.

Source: settings.py balloons_drag / balloons_require_jam_decoy; pybullet_balloons_base.py chute and attachment geometry.

THE SCENE

The box must pass between two walls

Mechanism schematic, not a recorded trajectory



- The slot is **9.6 cm wide**, from z = 0.50 to 1.00 m.
- Attachment offsets span **−2.8 to +2.8 cm**, ordered by clip index.
- Walls collide with **the box only**. The arm and balloons pass through them.
- Balanced net pull alone is not a guarantee: the full motion and contacts determine whether the box clears the slot.

Source: pybullet_balloons_base.py, chute_half_gap=0.048, chute_z_lo/hi, attach_span=0.028 and collision filters.

Same objects and skills, richer motion

Object	Observed features	Role
Box	`x, y, z, color, speed`	Material identity, position and linear speed; box orientation is absent from this feature vector.
Balloon	`x, y, z, color, tied, popped`	Gas identity; released/attached status; burst status.
Clip	`x, y, z, rot, is_on`	Clip i releases balloon i; an open clip stays open.
Band	`x, y, lo, hi`	Acceptable box-centre heights, 5 cm apart.
Robot	`x, y, z, fingers, roll, tilt, wrist`	End-effector and gripper state.

Release(robot, clip)

Push the toggle open. Parameters set approach distance and contact height. A freed balloon cannot be clipped back.

Wait(robot)

Hold still while the physics evolves. A jam is not automatically a terminal loss; it can leave the task unsolved.

Source: pybullet_balloons_base.py object types; ground_truth_models/balloons/options.py; sweep observations are noiseless.

Height is only the free-ascent prediction

$$L_c(z) = L_c \cdot \max(0, 1 - (z - z_{table}) / h)$$

$$z^* = z_{table} + h \cdot (1 - W / \Sigma L_c)$$

W includes the box and freed balloons. The second expression assumes an unconstrained equilibrium; it does not model overshoot or wall contact.

Quantity	Sweep value	Meaning
Red / blue / green / gold lift	0.35 / 0.50 / 0.70 / 1.00 N	Per-colour lift at table height
Fade height	0.80 m	Lift declines with height
Pine / oak box mass	0.05 / 0.08 kg	Both materials occur in the sweep
Balloon mass / air drag	0.005 kg / 2.2	Freed mass and transient damping
Ceiling / settle threshold	z = 1.18 m / speed < 0.01 m/s	Burst boundary and the evaluator's low-speed check

True constants shown for explanation, not supplied as answers to the agent. Source: settings.py and pybullet_balloons.py hover_height().

Test levels include a near-height jamming decoy

1. Draw a rack and box material; enumerate balloon subsets and their unconstrained hover heights.
2. Choose a 5 cm band within the chute and simulate the in-band candidates. Keep a level with **one safe subset** under this check.
3. For test, require the candidate **nearest the band centre** to fail to settle in-band **without bursting**: the jamming decoy.
4. Its predicted hover height must be within **2 cm** of the safe subset's. Check that the oracle's actual release plan succeeds.

> Height alone can favour the wrong subset. Release order and the path through the chute still matter during execution.

The generation probe opens the candidate clips and rolls forward for up to 400 steps. A separate oracle check validates a sequence of Release / Wait skills.

Source: pybullet_balloons.py _make_tasks() / subset_outcome(); balloons_require_jam_decoy=True, contact_height_tol=0.02, max_sampling_attempts=80.

Two near-identical height predictions

Oak box; rack order: green, red, blue, gold.
Recorded target band: **[0.5023, 0.5523] m**.

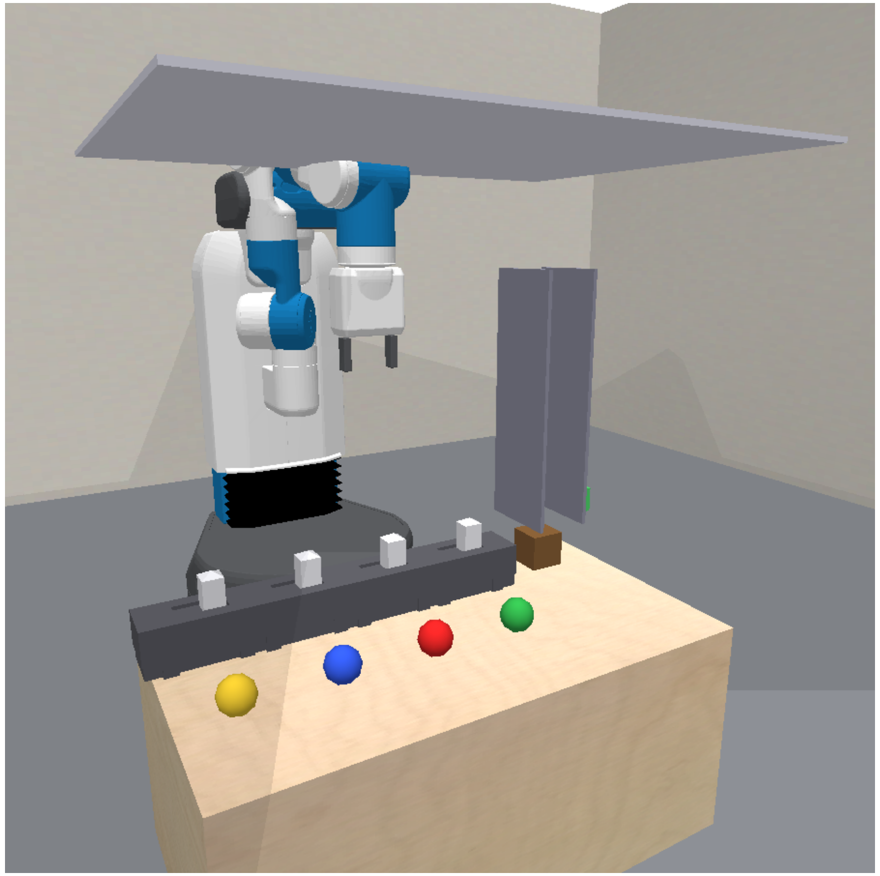
Subset	Total lift	Analytic z*	Interpretation
Gold (clip 3)	1.00 N	0.5329 m	In band; the recorded agent released it and won.
Green + red (clips 0, 1)	1.05 N	0.5273 m	Also in band, closer to its centre; height cannot certify it.

The predictions differ by only **5.6 mm**. The old isolated-hover-height example no longer describes this task.

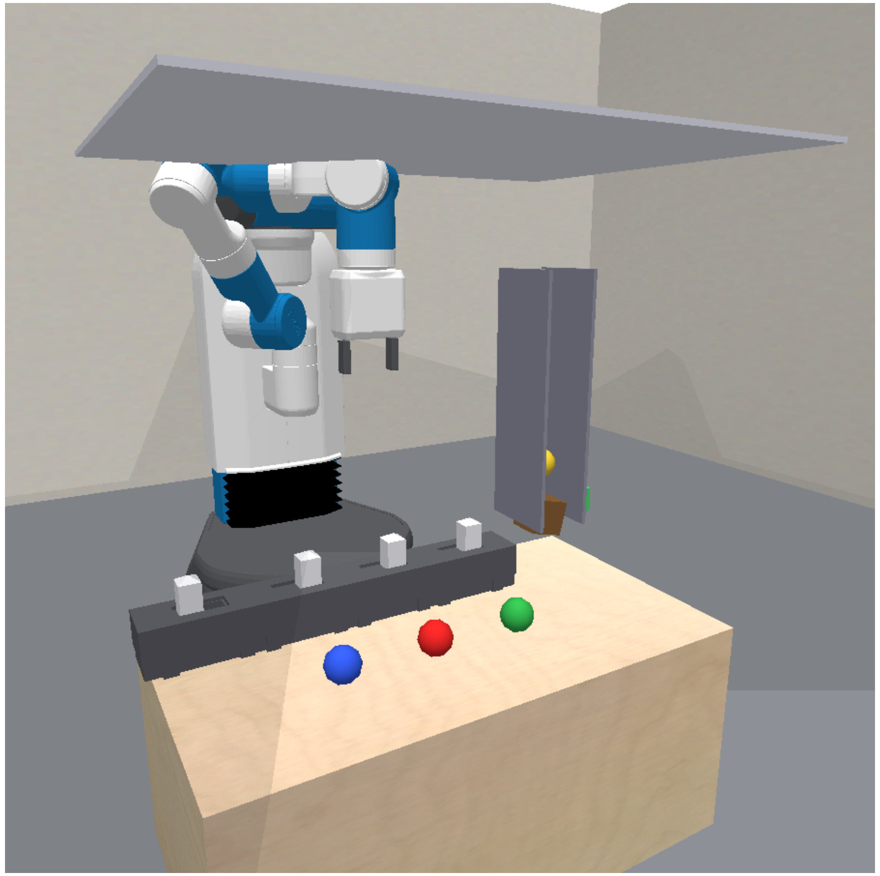
The successful real sequence was Release(clip3), 41 steps, then Wait, 7 steps. The alternative's height above is calculated from the source law, not a newly replayed counterfactual.

Source: uniform MB seed 0, run_20260907_102820, L03/index.jsonl and level observation; analytic values use the true source constants.

Gold alone solves this test level



Before: four clipped balloons, box below the chute.



After: gold released; the evaluator reports WIN at test step 48.

The chute partially hides the box and green band in this camera view. These are preserved experiment frames, not a replay on newer code.

Source: balloons-agent_continual_uniform/seed0/run_20260907_102820/L03/renders/{ep000_start,ep000_win}.png.

Three levels per seed, one continuing agent

	Training	Test
Levels per seed	2	1
Balloons per level	2 or 3	4, the full palette
Box material	Pine or oak	Pine or oak
Resets	Allowed and counted	Unavailable

- Training draws prefer colours and materials not yet shown; generation can fall back to a free draw.
- The journal, learned model and recorded data carry forward to each new level. Test composes familiar identities in a new rack.
- The sweep uses seeds 0–2 and noiseless observations. Report solve rate, total real steps and agent resets over **all train + test levels**.

Source: protocol_continual_uniform_sweep.yaml and saved scorecards; 5,000 steps per level pooled into a 15,000-step run cap, 48-hour wall-clock cap.

What the simulator adds

Model-based (MB)

- Uses a visible base simulator with the scene, chute geometry and rigid-body contacts.
- Writes residual rules for release, attachment, lift and bursting; fits uncertain physical parameters.
- Can test release plans in its learned simulator before paying real steps.

Model-free baseline (MF)

- Has the same real environment, skill tools, observations, journal and recorded experience.
- Has no supplied simulator or fit-and-rollout workbench.
- Still has arbitrary Python: it can fit equations and reason about dynamics by hand.

> Contact prediction is the intended benefit. The baseline's name does not mean it must guess or cannot build an analytic model.

Source: uniform-sweep arm configuration and balloons agent tool contract. Sandbox computation does not count as real environment steps.

COMPLETED UNIFORM SWEEP

Balloons: a modest gap in these three seeds

Arm	Seed	Levels solved	Total real steps	Resets
MB	0	3 / 3	273	0
MB	1	3 / 3	385	1
MB	2	3 / 3	787	1
MB	**Mean**	**100.0%**	**481.7**	**0.67**
MF	0	3 / 3	506	1
MF	1	2 / 3	558	1
MF	2	3 / 3	423	1
MF	**Mean**	**88.9%**	**495.7**	**1.00**

MB: **9/9** levels; MF: **8/9**. Mean step counts are close (482 vs 496). These runs do not show a large balloons step-efficiency advantage.

Means cover all three levels, including unfinished levels in completed unsuccessful runs. Steps exclude sandbox rollouts; lower counts can reflect stopping before solving.

Source: six completed *_uniform scorecards pinned in sweep_20260907/manifest.json. These are the uniform sweep, not the earlier contact A/B results.

The deck now describes the sweep version

Source	What to inspect
`predicators/envs/pybullet_balloons_base.py`	Scene, chute collision filters, attachment offsets and observed features.
`predicators/envs/pybullet_balloons.py`	Lift, attachment, burst, task generation, oracle checks and evaluator.
`predicators/settings.py`	Drag 2.2; contact-decoy default; 2 cm tolerance; training palette.
`protocol_continual_uniform_sweep.yaml`	Two training levels, common MB/MF flags and noiseless protocol.
`docs/envs/balloons/sweep_20260907/`	Saved screenshots, source manifest, per-seed results and hashes.
`docs/slides/make_balloons_domain_slides.py`	Rebuilds both standalone HTML decks; --pdf also renders the PDF.

> The older September 6 oracle video and stills show the original environment. They are retained as historical assets and are not used in this deck.